

# Venkata Weekshith

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## Profile

B.Tech Electrical Engineering student at IIT Bhubaneswar with hands-on experience in **machine learning, deep learning and signal processing** through EEG-iEEG research. Experienced in **Python, C++, PyTorch, NumPy, SciPy, MNE-Python and MATLAB**, with work involving synthetic medical-signal generation, event detection, spectral analysis and VAE-cGAN based EEG-to-iEEG reconstruction. Strong foundation in **data structures, algorithms, problem solving and software development**.

## Education

**Indian Institute of Technology Bhubaneswar**

**CGPA: 8.54/10**

B.Tech in Electrical Engineering

*Aug 2023 – Present*

**Sasi Educational Institute**

**94.4%**

Senior School Certificate Examination, CBSE

*Apr 2021 – Mar 2023*

## Experience

**Research Intern – Brain Signal Processing / EEG-iEEG Research Lab, IIT Bhubaneswar** *Jun 2026 – Jul 2026*

- Generated synthetic paired **20-channel EEG and 12-channel iEEG datasets for 18 subjects** using latent neural sources and source-to-sensor mixing for brain-signal analysis.
- Developed **IED and Non-IED event generation pipelines** with source-level signal injection and automated event-window extraction for signal analysis.
- Evaluated generated data using **correlation, power spectral density and inter-subject spectral similarity** to assess signal realism and consistency.
- Implemented and evaluated a **VAE-cGAN** architecture combining VAE, SPADE residual blocks, LSTM layers and PatchGAN for **EEG-to-iEEG reconstruction**.

## Projects

**SplitWise Clone – Expense Management Application**

*React.js, Node.js, Express.js, PostgreSQL, JWT*

GitHub | Live Demo

- Built a full-stack application with **JWT authentication**, protected routes, authorization middleware and REST APIs.
- Designed group and expense management workflows supporting **member validation, equal/custom expense splitting and accurate paise-level calculations**.
- Designed a relational **PostgreSQL schema** using primary/foreign keys and junction tables for users, groups, expenses and settlements.
- Implemented balance calculation and a **settlement state machine** supporting pending, accepted/rejected requests and settlement history.
- Deployed the complete application on **Render**, integrating frontend and backend services through REST APIs.

**Placement Checklist**

*React.js, JavaScript, HTML, CSS*

GitHub | Live Demo

- Built a React-based preparation dashboard for organizing **DSA, React, Core CS and Aptitude** preparation.
- Implemented task creation, completion tracking, priority levels, category filtering and real-time progress statistics.
- Developed a responsive interface and deployed the application using **GitHub Pages**.

## Technical Skills

**Languages:** C++, Python, JavaScript, SQL

**CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS

**ML / Signal Processing:** PyTorch, NumPy, Pandas, SciPy, MNE-Python, MATLAB

**Backend:** Node.js, Express.js, REST APIs, JWT

**Frontend:** HTML, CSS, React.js

**Databases:** PostgreSQL, SQLite

**Tools:** Git, GitHub, VS Code, Google Colab, Render

## Certifications

**NxtWave:** Python | JavaScript | Node.js & Express.js | Responsive Web Development

## Positions of Responsibility

**Core Team Member – Pravaah, IIT Bhubaneswar**

*Sep 2024 – May 2025*

- Delegated tasks to junior members and volunteers while coordinating event responsibilities.
- Negotiated sponsorship deliverables, sponsorship tiers and monetary/in-kind contributions.

**Associate – E-Summit, IIT Bhubaneswar**

*Oct 2023 – Sep 2024*

- Collaborated with the core team to execute sponsor deliverables including branding, social media mentions and on-site visibility.